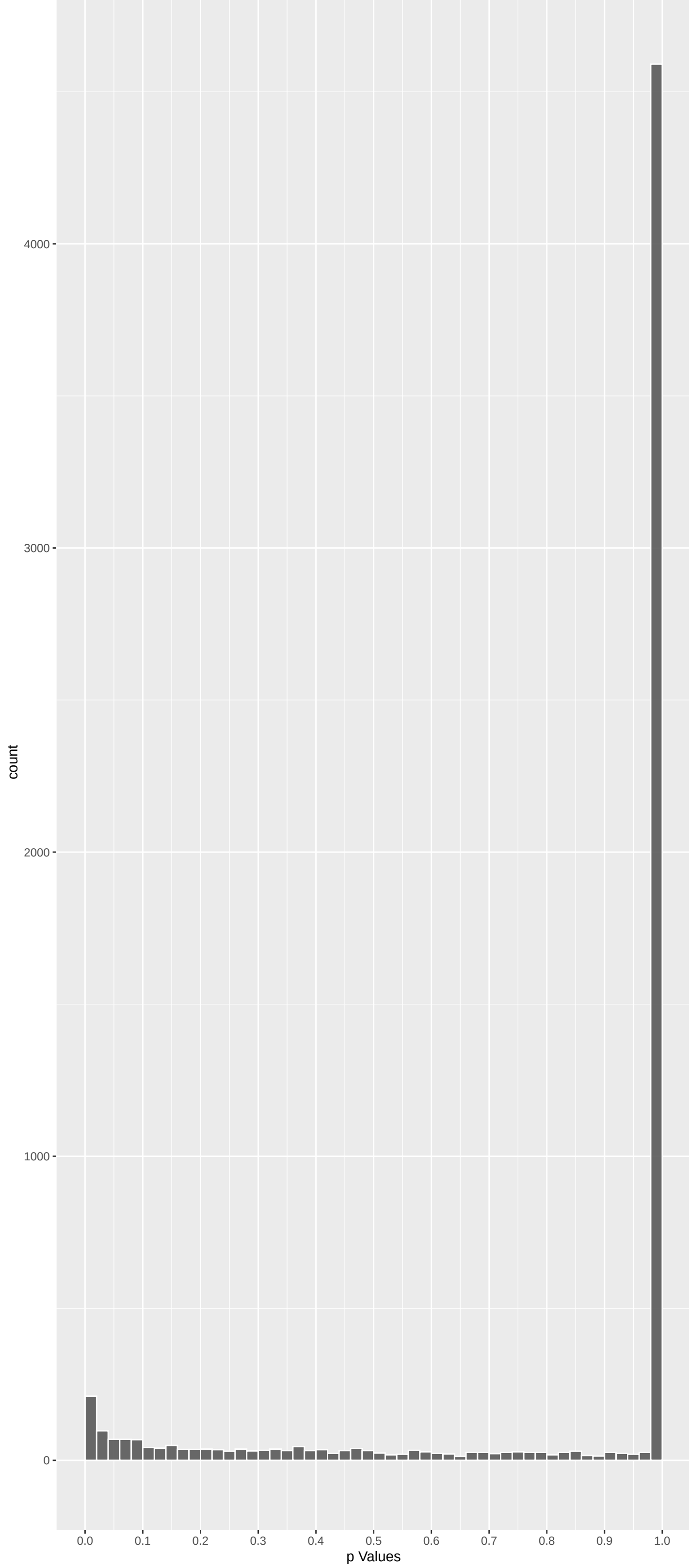


Top genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
ZSCAN5B	6.594745	2.555929e-10	4.081e-06	6.516e-02
ZSCAN5C	6.359737	1.212594e-09	9.681e-06	7.729e-02
TRANK1	-6.207586	3.228275e-09	1.718e-05	9.145e-02
CAMSAP1	-5.907006	2.090289e-08	8.344e-05	2.837e-01
MROH2A	-5.798872	4.005750e-08	1.066e-04	2.837e-01
PCDH15	-5.827026	3.385435e-08	1.066e-04	2.837e-01
ATP6V0A2	-5.681227	8.023913e-08	1.601e-04	3.054e-01
DYNC2H1	-5.699942	7.190893e-08	1.601e-04	3.054e-01
PKD1L1	-5.648623	9.704267e-08	1.722e-04	3.054e-01
OR6B1	-5.546423	1.749441e-07	2.793e-04	4.147e-01
IQANK1	-5.476135	2.608298e-07	3.471e-04	4.147e-01
RIMBP2	-5.477128	2.593708e-07	3.471e-04	4.147e-01
LZTFL1	-5.458202	2.885879e-07	3.545e-04	4.147e-01
POLD2	-5.440495	3.187962e-07	3.636e-04	4.147e-01
FBXW12	5.413673	3.704688e-07	3.944e-04	4.198e-01
SSTR4	-5.374653	4.603824e-07	4.594e-04	4.386e-01
ZBTB21	-5.360770	4.972090e-07	4.670e-04	4.386e-01
ATP2A1	-5.190985	1.255104e-06	1.113e-03	9.876e-01
ZFYVE9	5.157155	1.504383e-06	1.264e-03	9.999e-01
TRPM5	-5.146894	1.589011e-06	1.269e-03	9.999e-01
NAA16	5.130967	1.729546e-06	1.315e-03	9.999e-01
FBXO24	-5.106858	1.965358e-06	1.426e-03	1.000e+00
CHRN2	-5.067972	2.412465e-06	1.605e-03	1.000e+00
MYH13	-5.073181	2.347324e-06	1.605e-03	1.000e+00
CEMP	-5.043369	2.744436e-06	1.753e-03	1.000e+00

Positive Rho Non-permuted



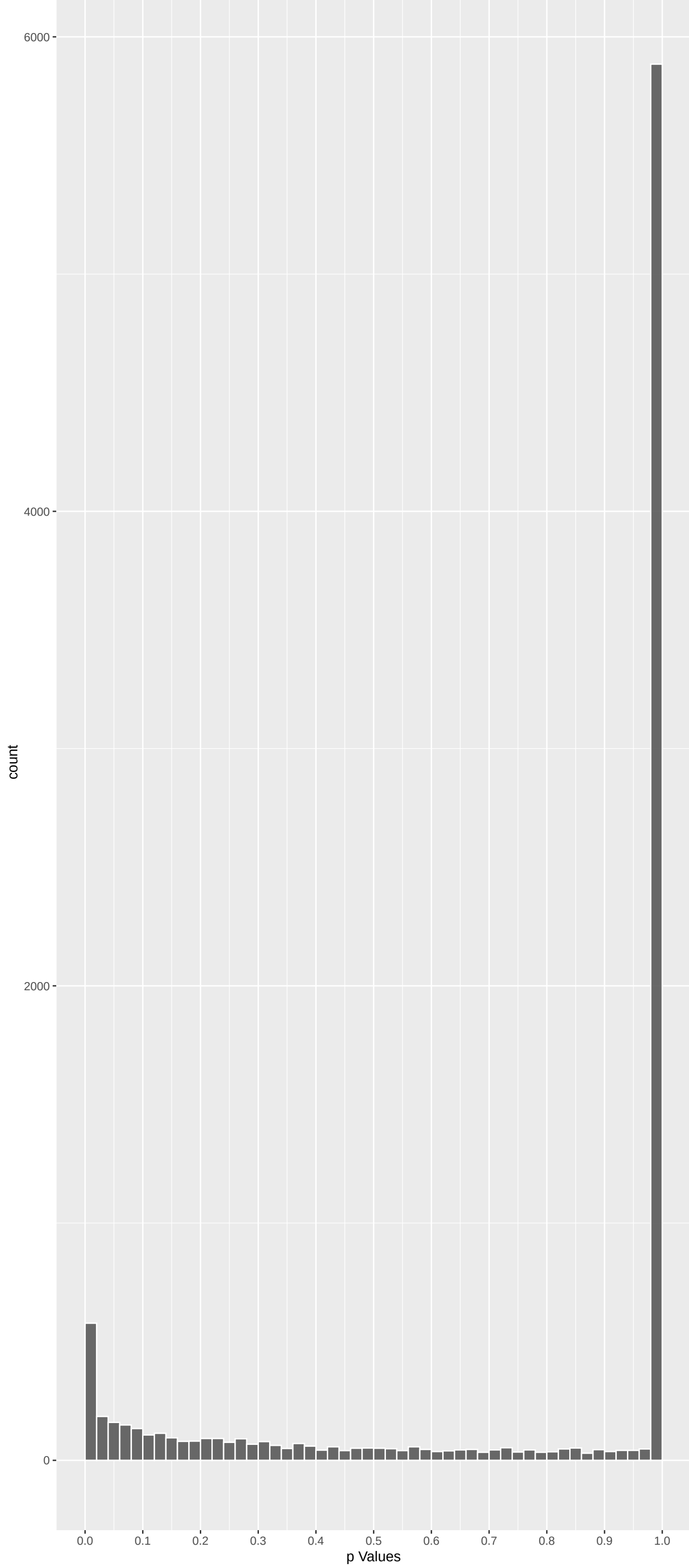
Top Positive genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
ZSCAN5B	6.594745	2.555929e-10	4.081e-06	6.516e-02
ZSCAN5C	6.359737	1.212594e-09	9.681e-06	7.729e-02
FBXW12	5.413673	3.704688e-07	3.944e-04	4.198e-01
ZFYVE9	5.157155	1.504383e-06	1.264e-03	9.999e-01
NAA16	5.130967	1.729546e-06	1.315e-03	9.999e-01
CATSPERE	5.018466	3.125149e-06	1.898e-03	1.000e+00
ATP2B4	4.929272	4.952189e-06	2.259e-03	1.000e+00
UBR2	4.831569	8.127669e-06	3.507e-03	1.000e+00
CATSPERB	4.741795	1.271000e-05	4.832e-03	1.000e+00
ZSCAN20	4.749304	1.224710e-05	4.832e-03	1.000e+00

Top genes by Q-Value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
ZSCAN5B	6.5947447	2.555929e-10	4.081e-06	6.516e-02
ZSCAN5C	6.3597373	1.212594e-09	9.681e-06	7.729e-02
TRANK1	-6.2075862	3.228275e-09	1.718e-05	9.145e-02
CAMSAP1	-5.9070058	2.090289e-08	8.344e-05	2.837e-01
MROH2A	-5.7988719	4.005750e-08	1.066e-04	2.837e-01
PCDH15	-5.8270257	3.385435e-08	1.066e-04	2.837e-01
PKD1L1	-5.6486234	9.704267e-08	1.722e-04	3.054e-01
ATP6V0A2	-5.6812270	8.023913e-08	1.601e-04	3.054e-01
DYNC2H1	-5.6999420	7.190893e-08	1.601e-04	3.054e-01
LZTFL1	-5.4582022	2.885879e-07	3.545e-04	4.147e-01
IQANK1	-5.4761349	2.608298e-07	3.471e-04	4.147e-01
POLD2	-5.4404951	3.187962e-07	3.636e-04	4.147e-01
OR6B1	-5.5464226	1.749441e-07	2.793e-04	4.147e-01
RIMBP2	-5.4771280	2.593708e-07	3.471e-04	4.147e-01
FBXW12	5.4136732	3.704688e-07	3.944e-04	4.198e-01
SSTR4	-5.3746528	4.603824e-07	4.594e-04	4.386e-01
ZBTB21	-5.3607696	4.972090e-07	4.670e-04	4.386e-01
ATP2A1	-5.1909854	1.255104e-06	1.113e-03	9.876e-01
ZFYVE9	5.1571548	1.504383e-06	1.264e-03	9.999e-01
NAA16	5.1309669	1.729546e-06	1.315e-03	9.999e-01
TRPM5	-5.1468938	1.589011e-06	1.269e-03	9.999e-01
M6PR	-2.6426342	4.935832e-02	6.702e-01	1.000e+00
FKBP4	-0.7742401	1.000000e+00	1.000e+00	1.000e+00
CYP26B1	-2.3460078	1.138543e-01	9.961e-01	1.000e+00
NDUFAF7	-0.1581565	1.000000e+00	1.000e+00	1.000e+00

Negative Rho Non-permuted



Top Negative genes by P-value non-permuted

Gene	Rho	P	p.adj	qValueNoperm
TRANK1	-6.207586	3.228275e-09	1.718e-05	9.145e-02
CAMSAP1	-5.907006	2.090289e-08	8.344e-05	2.837e-01
MROH2A	-5.798872	4.005750e-08	1.066e-04	2.837e-01
PCDH15	-5.827026	3.385435e-08	1.066e-04	2.837e-01
ATP6V0A2	-5.681227	8.023913e-08	1.601e-04	3.054e-01
DYNC2H1	-5.699942	7.190893e-08	1.601e-04	3.054e-01
PKD1L1	-5.648623	9.704267e-08	1.722e-04	3.054e-01
OR6B1	-5.546423	1.749441e-07	2.793e-04	4.147e-01
IQANK1	-5.476135	2.608298e-07	3.471e-04	4.147e-01
RIMBP2	-5.477128	2.593708e-07	3.471e-04	4.147e-01

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Atrial Cardiac Muscle Cell Action Potent	-0.36823663	11	2.351e-05	6.333e-02	CACNB2:58 GJC1:243 SCN5A:536 KCNN2:896 GJA1:1171 KCNE5:1480
Cardiac Muscle Cell Action Potential Inv	-0.23434356	28	1.782e-05	6.333e-02	KCNH2:189 GJC1:243 CACNA1G:461 SCN5A:536 KCNE2:632 GJA1:1171
NADH Dehydrogenase Complex Assembly (GO:016169979)	0.16169979	48	1.074e-04	7.236e-02	NDUFA11:113 NDUFB8:126 NDUFAF1:134 NDUFS5:238 OXAL1:251 NDUFS3:301
Mitochondrial Respiratory Chain Complex	0.16169979	48	1.074e-04	7.236e-02	NDUFA11:113 NDUFB8:126 NDUFAF1:134 NDUFS5:238 OXAL1:251 NDUFS3:301
Positive Regulation Of Tyrosine Phosphor	0.16341999	49	7.661e-05	7.236e-02	IL6ST:71 GH2:179 PARP9:221 IL4:226 CSF1R:243 CSH2:255
Potassium Ion Transmembrane Transport (G	-0.10283296	132	4.686e-05	7.236e-02	KCN4A:68 KCNF1:115 KCNV2:156 KCNH2:189 KCNK18:196 SLC12A4:199
Regulation Of Cardiac Muscle Cell Membra	-0.25179390	20	9.720e-05	7.236e-02	KCNH2:189 SCN5A:536 KCNE2:632 NOS1AP:886 KCNE5:1480 SNTA1:1551
Regulation Of Heart Rate By Cardiac Cond	-0.18757818	38	6.353e-05	7.236e-02	CACNB2:58 KCNH2:189 CACNA1G:461 SCN5A:536 KCNE2:632 KCNK2:1022
Regulation Of Ventricular Cardiac Muscle	-0.25754099	17	2.371e-04	1.420e-01	KCNH2:189 SCN5A:536 KCNE2:632 NOS1AP:886 KCNE5:1480 SNTA1:1551
Oxidative Phosphorylation (GO:0006119)	0.14195122	55	2.741e-04	1.477e-01	NDUFA11:113 NDUFB8:126 NDUFS5:238 NDUFS3:301 ATP5PD:425 NDUFA12:723
Potassium Ion Transport (GO:0006813)	-0.09573849	119	3.178e-04	1.556e-01	KCN4A:68 KCNF1:115 KCNV2:156 KCNH2:189 KCNK18:196 SLC12A4:199
Cardiac Conduction (GO:0061337)	-0.15636386	43	3.917e-04	1.623e-01	CACNB2:58 KCNH2:189 CACNA1G:461 SCN5A:536 KCNE2:632 KCNK2:1022
Proton Motive Force-Driven Mitochondrial	0.15037531	7	3.648e-04	1.623e-01	NDUFA11:113 NDUFB8:126 NDUFS5:238 NDUFS3:301 ATP5PD:425 NDUFA12:723
Potassium Ion Export Across Plasma Membr	-0.29074855	12	4.882e-04	1.754e-01	KCNH2:189 KCNK18:196 KCNE2:632 KCND3:1022 KCNE5:1480 KCNE1:1691
Ventricular Cardiac Muscle Cell Action P	-0.25304333	16	4.585e-04	1.754e-01	KCNH2:189 SCN5A:536 KCNE2:632 KCNE5:1480 SNTA1:1551 NEDD4L:1603
Atrial Cardiac Muscle Cell To AV Node Ce	-0.40231686	6	6.430e-04	2.165e-01	GJC1:243 SCN5A:536 GJA1:1171 KCNE5:1480 SCN3B:2699 KCNK1:3457
Mitochondrial Electron Transport, NADH T	0.17567919	31	7.143e-04	2.264e-01	NDUFB8:126 NDUFAF1:134 NDUFS5:238 NDUFS3:301 NDUFB5:783 NDUFB10:1188
Cellular Respiration (GO:0004533)	0.11556861	71	7.694e-04	2.303e-01	NDUFA11:113 NDUFB8:126 NDUFS5:238 OXAL1:251 NDUFS3:301 BLOC1C1S1:507
Aerobic Respiration (GO:0009060)	0.13175380	54	8.184e-04	2.321e-01	NDUFA11:113 NDUFB8:126 NDUFS5:238 OXAL1:251 NDUFS3:301 BLOC1C1S1:507
Chemical Synaptic Transmission (GO:00072	-0.06000401	253	1.067e-03	2.640e-01	RIMBP2:9 CHRN2:19 HTR3B:41 CHRNA3:48 CACNB2:58 GRIN3B:62
Monoatomic Cation Transmembrane Transpor	-0.05803821	273	1.019e-03	2.640e-01	ATP6V0A2:6 ATP2A1:15 TRPM5:16 KCNA4:68 KCNF1:115 KCNV2:156
Regulation Of Rho Protein Signal Transdu	-0.11476776	68	1.078e-03	2.640e-01	GPR35:99 EPS8L3:136 GPR20:247 EPS8L1:279 AKAP13:390 RIPOR2:440
Regulation Of Smooth Muscle Contraction	-0.20922291	20	1.201e-03	2.814e-01	CHRNA3:48 TACR2:121 PROK2:493 SETD3:942 CHRM3:1081 CTNN1:1395
Regulation Of Tyrosine Phosphorylation O	0.12365120	57	1.253e-03	2.814e-01	IL6ST:71 GH2:179 PARP9:221 IL4:226 CSF1R:243 CSH2:255
Neuropeptide Signaling Pathway (GO:00072	-0.11716023	63	1.313e-03	2.830e-01	SSTR4:13 PENK:270 OPRK1:367 NTSR2:152 RXFP3:737 OPR1L:848
Cardiac Muscle Cell Action Potential (GO:	-0.10780375	29	1.460e-03	3.026e-01	KCNH2:189 CACNA1G:461 SCN5A:536 KCNE2:632 KCND3:1022 HCN2:1177
Regulatory ncRNA Processing (GO:0070918)	-0.16194314	32	1.529e-03	3.050e-01	MOV10L1:104 TDRKH:129 PIWIL1:195 ANKRD34C:264 MAEL:680 ANKRD34B:1025
Proton Motive Force-Driven ATP Synthesis	0.12689437	51	1.732e-03	3.334e-01	NDUFA11:113 NDUFB8:126 NDUFS5:238 NDUFS3:301 ATP5PD:425 NDUFA12:723
Intermediate Filament Organization (GO:0	-0.12718007	50	1.878e-03	3.373e-01	KRT15:222 KRT19:327 KRT78:395 KRT80:602 KRT36:730 KRT35:825
piRNA Processing (GO:0034537)	-0.23194938	18	1.854e-03	3.373e-01	MOV10L1:104 TDRKH:129 PIWIL1:195 ANKRD34C:264 MAEL:680 ANKRD34B:1025
Ventricular Cardiac Muscle Cell Membrane	-0.33207627	7	2.346e-03	4.077e-01	KCNH2:189 KCNK2:632 KCNK3:1527 KCNE1:1691 KCNJ5:2499 KCNQ1:3457
Anterograde Trans-Synaptic Signaling (GO	-0.09648271	180	2.717e-03	4.351e-01	CHRN2:19 HTR3B:41 CHRNA3:48 CACNB2:58 HTR1D:75 OMP:77
Mitochondrial Respiratory Chain Complex	0.09883365	77	2.746e-03	4.351e-01	NDUFA11:113 NDUFB8:126 NDUFAF1:134 NDUFS5:238 OXAL1:251 NDUFS3:301
RNA Methylation (GO:0030488)	0.14461673	36	2.688e-03	4.351e-01	TRMT13:300 TRMT10A:313 BCDIN3D:386 TRMO:499 THUMP2D:502 TRMT6:527
Regulation Of Dendrite Development (GO:0	-0.13940076	38	2.958e-03	4.553e-01	CHRN2:19 CHRNA3:48 MFS2A:50 COBL:378 RAB17:383 KCNC1:1187
G Protein-Coupled Receptor Signaling Pat	-0.12493053	47	3.064e-03	4.586e-01	HTR1D:75 OR10J5:124 AGTR2:469 PTGIR:515 HTR6:594 MTNR1A:808
Aerobic Electron Transport Chain (GO:001	0.11559099	54	3.325e-03	4.594e-01	NDUFB8:126 NDUFAF1:134 NDUFS5:238 NDUFS3:301 NDUFB5:783 COX6B1:791
Cardiac Muscle Cell Contraction (GO:0086	-0.19992940	18	3.324e-03	4.594e-01	GSN:237 CACNA1G:461 SCN5A:536 KCNE2:632 PIK3CA:1557 KCNE1:1691
Mitochondrial ATP Synthesis Coupled Elec	0.11475769	55	3.266e-03	4.594e-01	NDUFB8:126 NDUFAF1:134 NDUFS5:238 NDUFS3:301 NDUFA12:723 NDUFB5:783
Nitric Oxide Mediated Signal Transductio	-0.20455062	17	3.506e-03	4.702e-01	AGTR2:469 NEUROD1:488 CD36:1259 RILPL1:1688 NOS1:1957 DDAH2:2049

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
BENPORATH_ES_WITH_H3K27ME3	-0.05889188	976	6.811e-10	4.419e-06	CEMP1:20 SUSD4:31 PLEC:35 LY6H:53 ADRB1:57 ZBTB16:60
MARTENS_TRINITIN_RESPONSE_UP	-0.06874780	675	1.434e-09	4.651e-06	INHBB:47 LY6H:53 FGF8:61 GRIN3B:62 CEP170B:73 GPR35:99
NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO	-0.15109777	127	4.245e-09	9.181e-06	HELZ2:59 MTG2:171 ABHD168:212 FAM217B:224 SLC04A1:245 SYCP2:306
HAMAL_APOPTOSIS_VIA_TRAIL_UP	0.06987262	591	7.861e-09	1.275e-05	BAZ2B:34 PTPN13:53 MIS18BP1:68 ANKH01:82 QSER1:86 CENPC:93
DODD_NASOPHARYNGEAL_CARCIOMA_DN	0.04798771	124	3.013e-08	3.910e-05	TET1:24 CD80:39 QSER1:86 CDC7:89 URIL:91 PPRC1:101
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	-0.07616601	419	9.955e-08	9.274e-05	CHRN2:19 CHRNA3:48 ZBTB16:60 FGF8:61 GRIN3B:62 HTR1D:75
DACOSTA_UV_RESPONSE_VIA_ERCC3_DN	0.05552342	809	1.001e-07	9.274e-05	ZFYVE9:4 ATP2B4:7 UBR2:8 AKAP9:15 BAZ2B:34 FAM169A:36
BENPORATH_SUZ12_TARGETS	-0.04849790	912	8.536e-07	5.538e-04	CEMP1:20 SUSD4:31 PLEC:35 ITPKB:42 CHRNA3:48 ADRB1:57
MIKKELSEN_NPC_HCP_WITH_H3K27ME3	-0.07943135	328	8.204e-07	5.538e-04	LY6H:53 BNC1:130 SIPR5:154 PENK:270 DLGAP2:277 C1QL2:284
REACTOME_CLASS_A_1_RHODOPSIN_LIKE_RECEPT	-0.09685977	296	7.787e-07	5.538e-04	SSTR4:13 ADRB1:57 GPR143:63 HTR1D:75 GPR35:99 GPR132:100
WP_GPCRS_CLASS_A_RHODOPSINLIKE	-0.09685368	211	1.296e-06	7.646e-04	OR6B1:8 SSTR4:13 ADRB1:57 HTR1D:75 GPR35:99 P2RY4:135
FISCHER_DEAM_TARGETS	0.04070441	865	3.148e-06	1.702e-03	ORC1:30 DNMT3B:54 MIS18BP1:68 CDC7:89 CENPC:93 RTKN2:95
MEISSNER_BRAIN_HCP_WITH_H3K4ME3_AND_H3K2	-0.04330605	1019	3.666e-06	1.830e-03	CHRN2:19 CEMP1:20 MINAR1:27 ITPKB:42 INHBB:47 MFS2A:50
MIKKELSEN_MEF_HCP_WITH_H3K27ME3	-0.05704919	550	5.313e-06	2.462e-03	CHRN2:19 RASGRP2:23 CHRNA3:48 LY6H:53 GRIN3B:62 CPNWE:95
BENPORATH_EED_TARGETS	-0.04367894	917	8.792e-06	3.803e-03	CEMP1:20 SUSD4:31 PLEC:35 INHBB:47 ADRB1:57 ZBTB16:60
REACTOME_SIGNALING_BY_GPCR	-0.05154213	623	1.263e-05	5.122e-03	SSTR4:13 RASGRP2:23 ADRB1:57 GPR143:63 HTR1D:75 GPR35:99
JOHNSTONE_PARVUS_TARGETS_3_DN	0.04564997	766	1.976e-05	7.490e-03	UBAP2:66 MIS18BP1:68 SENP7:70 QSER1:86 KTRK2:195 NABP:97
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERAC	-0.07670211	261	2.078e-05	7.490e-03	SSTR4:13 CHRN2:19 CHRNA3:48 ADRB1:57 GRIN3B:62 HTR1D:75
BENPORATH_PRC2_TARGETS	-0.05169699	576	2.468e-05	7.625e-03	CEMP1:20 SUSD4:31 PLEC:35 ITPKB:42 CHRNA3:48 ADRB1:57
RODRIGUES_THYROID_CARCIOMA_POORLY_DIFFE	0.05130001	586	2.442e-05	7.625e-03	GOT1:60 IL6ST:71 ZNF644:105 APC:116 WDR76:136 HPS3:145
DACOSTA_UV_RESPONSE_VIA_ERCC3_COMMON_DN	0.05884521	441	2.447e-05	7.625e-03	ZFYVE9:4 UBR2:8 AKAP9:15 BAZ2B:34 FAM169A:36 VPS13A:48
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	-0.06709733	300	2.950e-05	8.531e-03	LY6H:53 FGF8:61 BNC1:130 PENK:270 DLGAP2:277 C1QL2:284
GEORGES_TARGETS_OF_MIR192_AND_MIR215	0.04353486	837	3.024e-05	8.531e-03	ORC1:30 DST:49 MIS18BP1:68 IL6ST:71 USP54:80 CDC7:89
REACTOME_SPERM_MOTILITY_AND_TAXES	0.42411889	8	3.263e-05	8.566e-03	CATSPERB:10 CATSPER2:19 CATSPERG:32 CATSPERD:124 KCNU1:127 CATSPERA:4683
KEGG_ALLOGRAFT_REJECTION	0.25570555	22	3.301e-05	8.566e-03	CD86:18 CD80:39 IL4:226 CD28:41 IL12B:554 IFNG:558
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	-0.16796433	47	6.811e-05	1.700e-02	NDUFB8:126 NDUFAF1:134 NDUFS5:238 NDUFS3:301 NDUFA12:723 NDUFB5:783
REACTOME_GPCR_LIGAND_BINDING	-0.05778646	398	8.077e-05	1.941e-02	SSTR4:13 ADRB1:57 GPR143:63 HTR1D:75 GPR35:99 GPR132:100
NIKOLSKY_BREAST_CANCER_21Q22_AMPLICON	-0.29774673	14	1.146e-04	2.656e-02	COL6A1:172 SPATC1L:226 COL6A2:278 POFUT2:335 COL18A1:405 DIP2A:417
MEISSNER_BRAIN_HCP_WITH_H3K27ME3	-0.06972794	257	1.230e-04	2.752e-02	CHRNA3:48 FGF8:61 GRIN3B:62 COL8A2:90 BNC1:130 ALX4:210
NIKOLSKY_BREAST_CANCER_1Q21_AMPLICON	-0.19210430	33	1.342e-04	2.796e-02	TDRKH:129 PMVK:482 PYG2:644 RORC:707 PKLR:828 ADAM15:837
NIKOLSKY_BREAST_CANCER_12Q24_AMPLICON	-0.28412760	15	1.390e-04	2.796e-02	SFSWAP:79 GALTNT9:219 EP400:398 GOLGA3:756 NOC4L:1145 DD551:1249
REACTOME_COMPLEX_I_BIOGENESIS	0.15880372	48	1.416e-04	2.796e-02	NDUFA11:113 NDUFB8:126 NDUFAF1:134 NDUFS5:238 NDUFS3:301 NDUFA12:723
PUJANA_BRCA1_PCC_NETWORK	0.03049273	1428	1.422e-04	2.796e-02	ORC1:30 IRAG2:35 TMEM131L:55 CDC7:89 CENPC:93 UBR5:104
NIKOLSKY_BREAST_CANCER_16Q24_AMPLICON	-0.16103940	46	1.582e-04	3.018e-02	PIEZO1:32 ANKRD11:92 CPNWE:95 SPATA2L:206 ZC3H18:215 CBRFA2T3:246
BIOCARTA_TH1TH2_PATHWAY	0.24835167	19	1.785e-04	3.309e-02	CD86:18 IL18R1:218 IL4:226 CD28:41 IL12B:554 IFNG:558
YOSHIMURA_MAPK8_TARGETS_UP	0.03731108	1102	1.846e-04	3.328e-02	SSTR4:13 CHRN2:19 ITPKB:42 HI-6:43 INHBB:47 ADRB1:57
REACTOME_FERTILIZATION	0.22882170	21	2.836e-04	4.842e-02	CATSPERB:10 CATSPER2:19 CATSPERG:32 CATSPERD:124 KCNU1:127 CDP:263
PUJANA_ATM_PCC_NETWORK	0.03144663	1205	2.810e-04	4.842e-02	UBR2:8 IRAG2:35 TMEM131L:55 CDC7:89 CENPC:93 APC:116
CHARAFE_BREAST_CANCER_LUMINAL_VS_MESENCH	-0.05271653	404	2.917e-04	4.853e-02	CAPN8:44 INHBB:47 PER2:72 FAM83H:109 MB:126 SPATA2L:206
BIOCARTA_LAIR_PATHWAY	0.26901306	15	3.094e-04	4.895e-02	VCAM1:125 KNG1:141 ICAM1:450 ITGA4:643 TNF:1046 ITGAL:1202

DisGeNET Top pathways by non-permutation

Long QT Syndrome	-0.14307558	58	1.660e-04	8.142e-01	KCNA4:68 KCNH2:189 CUZD1:331 MINK1:368 RPGR:509 DNM2:526
Specific learning disability	0.12057948	86	1.132e-04	8.142e-01	NSD1:41 TNF2:289 DGCR6:295 SPG1:439 POMK:520 LTPB4:553
Acidosis, Lactic	0.05094047	150	3.168e-02	7.909e-01	NDUFA11:113 NDUFAF1:134 LRPPRC:162 TANGO2:204 G6PC:213 HSD17B6:296
Aneurysm, Dissecting	-0.15189697	18	2.570e-02	7.909e-01	KCNK218:85 LOXL1:311 PLOD1:655 NOS3:2058 APOE:2117 MTHFR:2815
Angioedemas, Hereditary	0.15916457	16	2.753e-02	7.909e-01	CNG1:141 CAB:637 ACE:811 AZM:832 SERPINAS:1476 SERPINAA:1575
ATRIAL FIBRILLATION, FAMILIAL, 1 (disord	-0.14936336	20	2.078e-02	7.909e-01	SCN5A:536 KCNE2:632 KCNE1:1691 GATP5:2027 SCN2B:2258 NPPA:2588
Brain Injuries, Focal	0.08483396	59	2.432e-02	7.909e-01	TNFSF10:174 IL18R1:218 TLR1:327 CHRNA7:449 IL31RA:546 IL1R1:899
Congenital serpinous diarrhea, sodium typ	-0.24266426	7	2.642e-02	7.909e-01	SLC9A3:1153 SLC9A2:1767 SPINT2:2175 HGF:2543 CAV1:3333 SLC9A1:8951
Dermatitis, Atopic	0.03606425	375	1.716e-02	7.909e-01	SLCO6A1:37 ALDH1A1:57 VCAM1:125 DNMT1:155 CXCR2:212 IL18R1:218
Diabetes Mellitus, Insulin-Dependent	0.02385339	736	2.956e-02	7.909e-01	CUX2:16 CD86:18 SLC06A1:37 CD80:39 G6PC2:87 APC:116
Empyema, Pleural	0.31604773	6	7.340e-03	7.909e-01	SMC2:529 NFKBIE:719 TNFRSF6B:1126 SMC4:1221 NFKBIA:7017 NFKBIB:7017
EPILEPSY, PYRIDOXINE-DEPENDENT	0.19484906	11	2.525e-02	7.909e-01	RNF19A:631 ALDH7A1:737 PLPBP:1508 CRK1:1612 GRAP2:7017 MAPK14:7017
FANCONI ANEMIA, COMPLEMENTATION GROUP A	0.04604449	214	1.972e-02	7.909e-01	APP:14 RNF168:46 TET2:50 ALDH1A1:57 BRP1:109 MEV1:118
HYPERTRICHOSIS, CONGENITAL GENERALIZED	0.17354073	14	2.457e-02	7.909e-01	IGF1:377 FGFR1:382 ABCB9:398 BRD2:819 BRCA1:1157 SOX3:7017
Ichthyosis, X-Linked	0.08977444	82	5.002e-03	7.909e-01	TNFSF10:174 ABCB1:177 IGF1:377 EGF:522 MDMA:564 KDR:709
Immunodeficiency syndrome, variable	0.36721620	5	4.458e-03	7.909e-01	DNMT3B:54 CDCA7:766 ATRX:1186 ZBTB24:1600 HELLS:7017 NA
Infant, Premature	0.04636799	216	1.926e-02	7.909e-01	SLCO6A1:37 DNMT3B:54 MBD1:79 LCT:83 ABCB1:177 TSLP:273
Injuries, Acute Brain	0.08483396	59	2.432e-02	7.909e-01	TNFSF10:174 IL18R1:218 TLR1:327 CHRNA7:449 IL31RA:546 IL1R1:899
Leprosy, Lepromatous	0.12408984	33	1.366e-02	7.909e-01	CD80:39 CD28:41 ICAM1:450 FOXP3:469 IL12B:554 IL1R1:899
Leukemia, Myelocytic, Acute	0.02195948	1365	7.917e-03	7.909e-01	CD86:18 TET1:24 SLC06A1:37 SLC22A16:38 CD80:39 NSD1:41
Lupus Erythematosus, Subacute Cutaneous	0.17516710	17	1.242e-02	7.909e-01	MBD1:79 DNMT1:155 CD28:41 MBD4:517 RBM45:1027 TNF:1046
Lupus Erythematosus, Systemic	0.03485716	844	6.998e-04	7.909e-01	CD86:18 SLC06A1:37 CD80:39 DNMT3B:54 PLA2R1:62 RTNK:295
Lymphoma, Non-Hodgkin	0.03763787	443	7.018e-03	7.909e-01	SLCO6A1:37 TET2:50 VCAM1:125 HDAC9:168 TNFSF10:174 ABCB1:177
Lymphoma, T-Cell, Cutaneous	0.05008112	210	1.268e-02	7.909e-01	CD80:39 STAT2:114 DNMT1:155 HDAC9:168 TNFSF10:174 ABCB1:177
Multiple Sclerosis, Primary Progressive	0.21676745	10	1.762e-02	7.909e-01	CSF1R:243 TLR:772 CSAP8:962 RBM45:1027 TNF:1046 LAMC2:7017
Neurogenic muscle atrophy, especially in	0.05251235	196	1.151e-02	7.909e-01	VPS13A:48 NDUFA11:113 COL2A1:129 NDUFA1:134 COL6A3:235 NEK1:278
Pain, Postoperative	0.15440044	21	1.433e-02	7.909e-01	TRPV1:165 ABCB1:177 SCN11A:388 ILF:813 TNF:1046 CACMK2B:1605
Psychoses, Drug	-0.19192014	20	2.971e-03	7.909e-01	CALR:180 DTNBP1:187 RGS9:292 ACT1:352 HTR6:594 CARMK2B:1605
Psychoses, Substance-Induced	-0.18622319	16	9.918e-03	7.909e-01	RGS9:292 ACT1:352 HTR6:594 CARMK2B:1605 PHB:1996 ESR1:2263
Schizophrenia, bipolar type	0.22456067	9	1.966e-02	7.909e-01	CHRNA7:449 DISC1:133 CD40:139 GABRA4:1549 NFKB1:7017 GABRB1:7017
Skin Diseases, Bullous	0.26331421	6	2.505e-02	7.909e-01	DST:49 COL7A1:96 RASA2:1497 RASA1:7017 IGAQ1:7017 DSG3:7017
Toxoplasmosis, Cerebral	0.32324781	6	6.077e-03	7.909e-01	IL6ST:71 LRPPRC:162 RBM45:1027 GATP:1639 IL6:7017 CD40LG:7017
Z1-hydroxyfolic acid deficiency	0.14346283	19	3.042e-02	7.909e-01	HSD17B6:296 CAA:542 CAB:637 BRD2:819 GLO1:920 HLA-DQA:1282
22q11 partial monosomy syndrome	0.19576300	11	2.457e-02	7.909e-01	PKAHA:708 JMDJ:986 TBX1:1039 HIRA:1557 COMT:7017 RREB1:7017
Abnormal mental state	0.24304083	8	1.729e-02	7.909e-01	SPG11:439 FMR1:887 CHL1:1659 GAB2:1793 CA2:7017 DYRK1A:7017
Abnormal mitochondria in muscle tissue	0.12402894	25	3.187e-02	7.909e-01	NDUFA11:113 NDUFAF1:134 NDUFS3:301 NDUFB1:1188 TMEM126B:1068 SDHA:7017
Abnormality of body height	-0.15248079	17	2.953e-02	7.909e-01	FGF8:61 DNMT1:369 PROKR2:493 GNHR1:533 TAC1:268 PROKR2:945
Abnormality of the outer ear	-0.09784543	45	2.309e-02	7.909e-01	PEX1A:198 KIAF1:591 FROK1:722 GATC1:1107 CCBC1:1109 GJA1:1171
Absence of pubertal development	-0.15350896	21	1.490e-02	7.909e-01	FGF8:61 WDR11:369 PROKR2:493 GNHR1:533 TAC1:268 PROKR2:945
Absent ear	0.24136986	7	2.704e-02	7.909e-01	ORC1:30 CHUK:303 CDC42:92 CDC45:7017 GMMN:7017 ORC6:7017